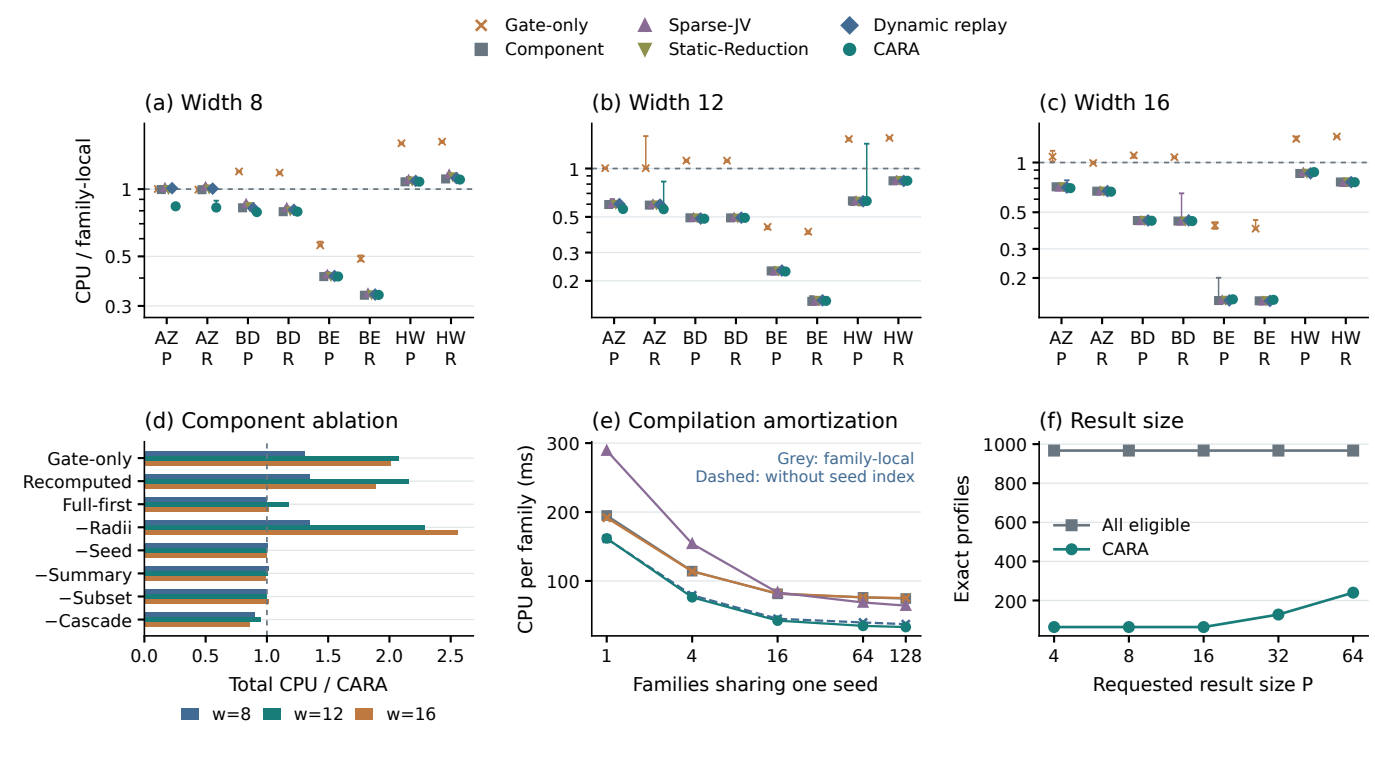
Gate-only
Component
Sparse-JV
Static-Reduction
Dynamic replay
CARA
(a) Width 8
(b) Width 12
(c) Width 16
CPU / family-local
1
0.5
0.3
0.2
AZ P
AZ R
BD P
BD R
BE P
BE R
HW P
HW R
(d) Component ablation
Gate-only
Recomputed
Full-first
−Radii
−Seed
−Summary
−Subset
−Cascade
0.0
0.5
1.0
1.5
2.0
2.5
Total CPU / CARA
w=8
w=12
w=16
(e) Compilation amortization
CPU per family (ms)
300
200
100
Grey: family-local
Dashed: without seed index
1
4
16
64
128
Families sharing one seed
(f) Result size
Exact profiles
1000
800
600
400
200
All eligible
CARA
4
8
16
32
64
Requested result size P